Gavin Kelley, ECGR 5411 Assignment 7

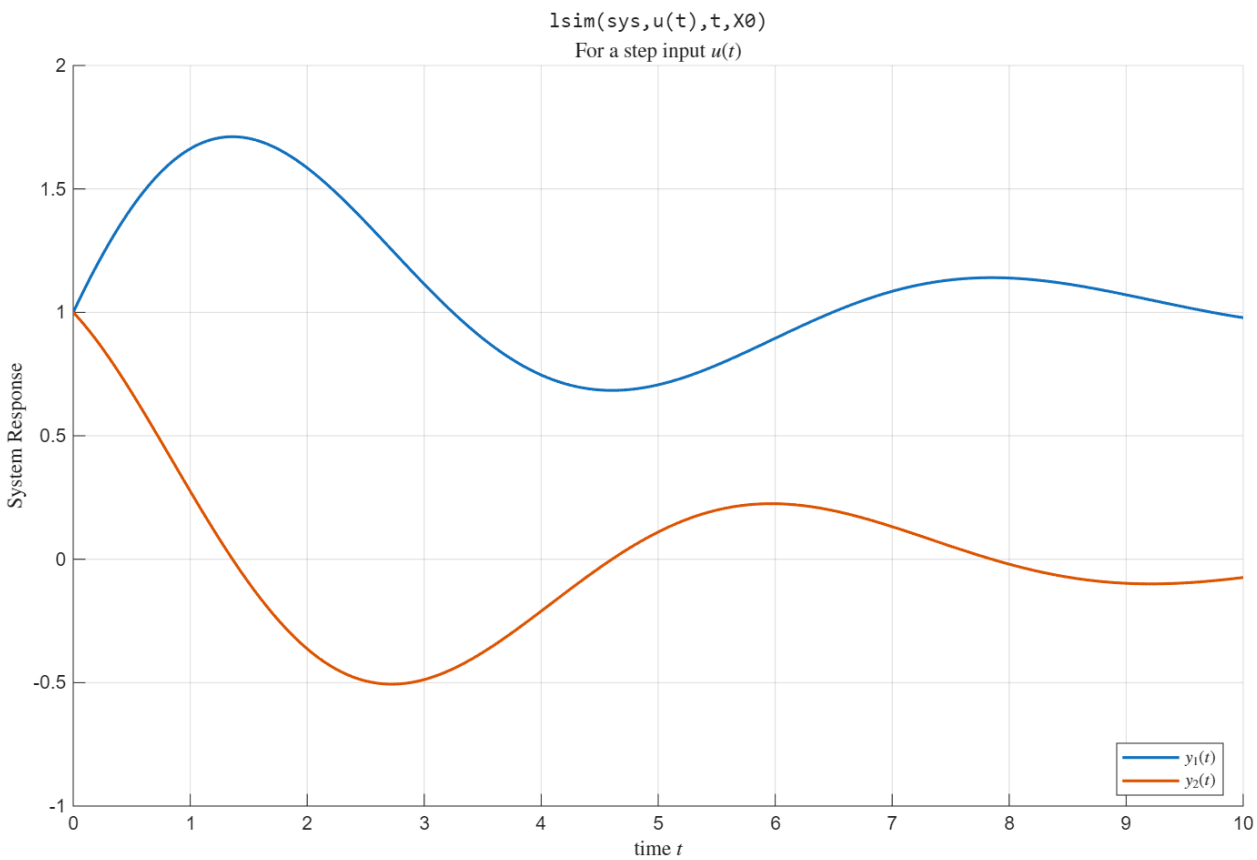
Problem 1

```
clc;clear;close all;
A = [0 1; -1 -0.5];
B = [0;1];
C = eye(2);
D = [0;0];

sys = ss(A,B,C,D);
```

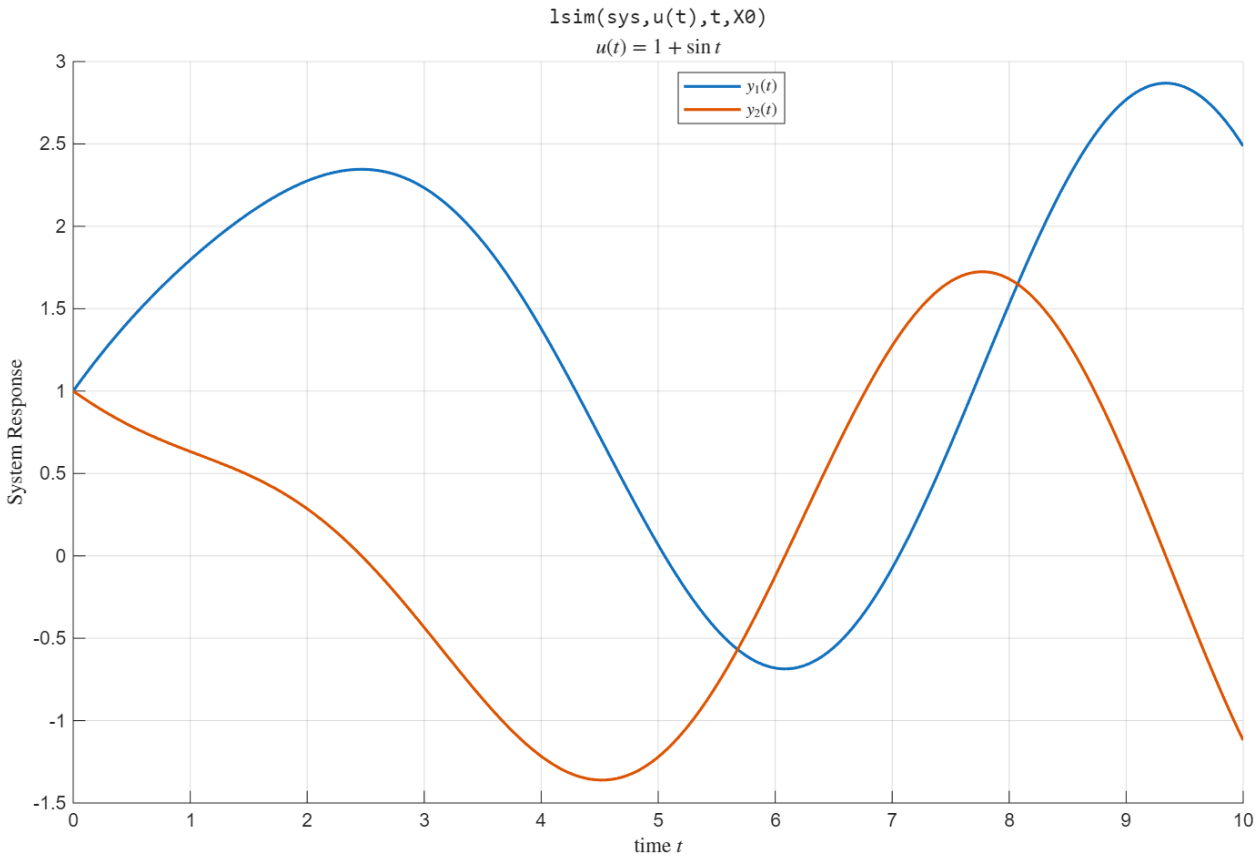
Problem 2

```
u=@(t) t>=0;
X0 = [1;1];
t = linspace(0,10,500);
[y, tout] = lsim(sys,u(t),t,X0);
figure; hold on; grid on;
plot(tout,y,'LineWidth',1.5);
xlabel('time $t$', 'Interpreter','latex');
ylabel('System Response', 'Interpreter','latex');
title('\texttt{lsim(sys,u(t),t,X0)}', 'Interpreter','latex');
subtitle('For a step input $u(t)$','Interpreter','latex');
legend({'$y_1(t)$','$y_2(t)$'},'Interpreter','latex','Location','best');
```



Problem 3

```
close all;
u=@(t) 1 + sin(t);
tspan = [0,10];
X0 = [1;1];
t = linspace(0,10,500);
[y, tout] = lsim(sys,u(t),t,X0);
figure; hold on; grid on;
plot(tout,y,'LineWidth',1.5);
xlabel('time $t$', 'Interpreter','latex');
ylabel('System Response', 'Interpreter','latex');
title('\texttt{lsim(sys,u(t),t,X0)}', 'Interpreter','latex');
subtitle('$u(t) = 1+\sin{t}$','Interpreter','latex');
legend({'$y_1(t)$','$y_2(t)$'},'Interpreter','latex','Location','best');
```

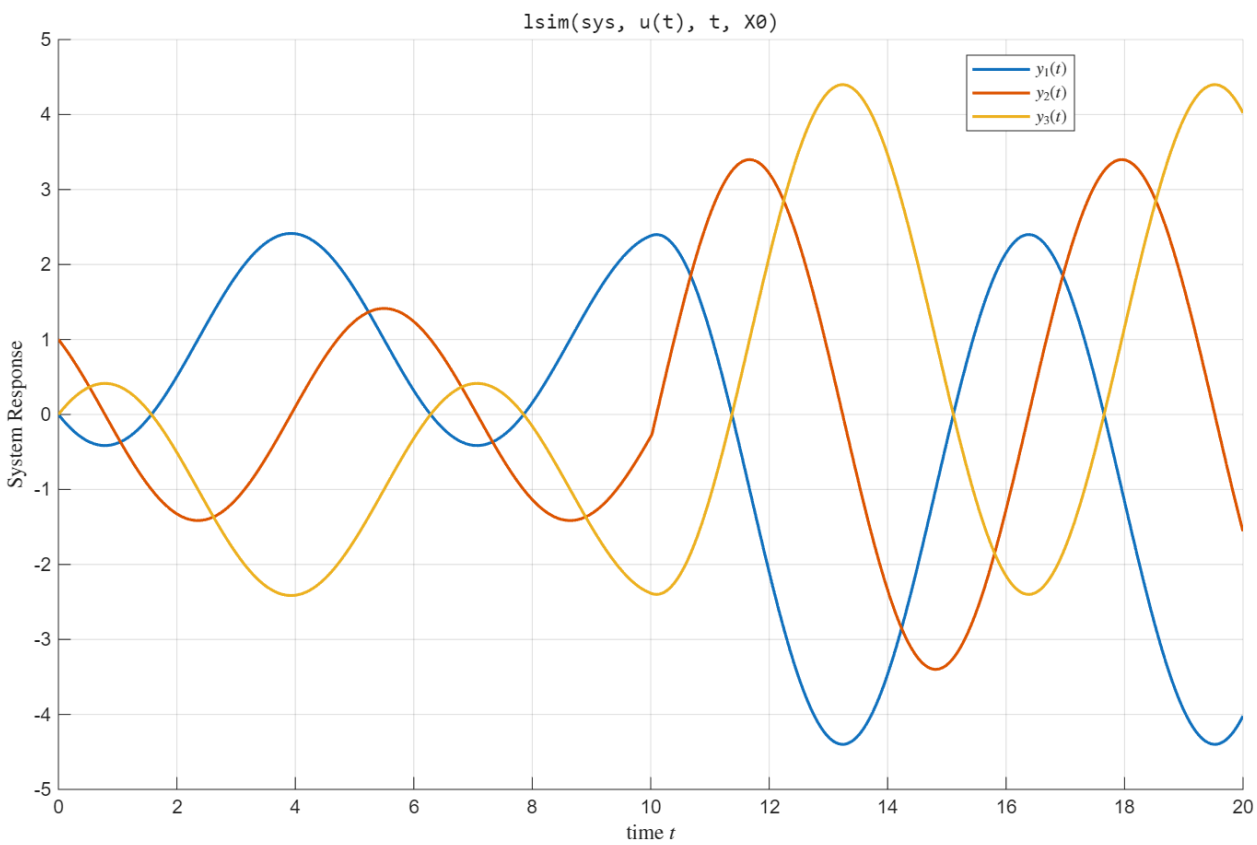


Problem 4

```
close all;
A = [0 1 -1 0; -1 0 0 0; 0 0 0 0; 0 -1 1 0];
B = [0;0;1;0];
C = [1 0 0 0; 0 -1 1 0; 0 0 0 1];
D = zeros(3,1);
X0 = [0;0;1;0];

% unit step function
step=@(t) t>=0;
% u(t) input
u=@(t) - ( step(t) - step(t-10) ) + ( step(t-10) - step(t-20) );
t = linspace(0,20,5000);

sys = ss(A,B,C,D);
[y, t_out, x] = lsim(sys, u(t), t, X0);
```

$$u(t) = \begin{cases} -1, & 0 \le t < 10, \\ 1, & 10 \le t < 20. \end{cases}$$

```
figure; hold on; grid on;
plot(t_out, y(:,1), 'LineWidth', 1.5);
plot(t_out, y(:,2), 'LineWidth', 1.5);
plot(t_out, y(:,3), 'LineWidth', 1.5);
title('\texttt{lsim(sys, u(t), t, X0)}','Interpreter','latex');
xlabel('time $t$', 'Interpreter','latex');
ylabel('System Response', 'Interpreter','latex');
legend({'$y_1(t)$','$y_2(t)$','$y_3(t)$'},'Interpreter','latex','Location','best');
```